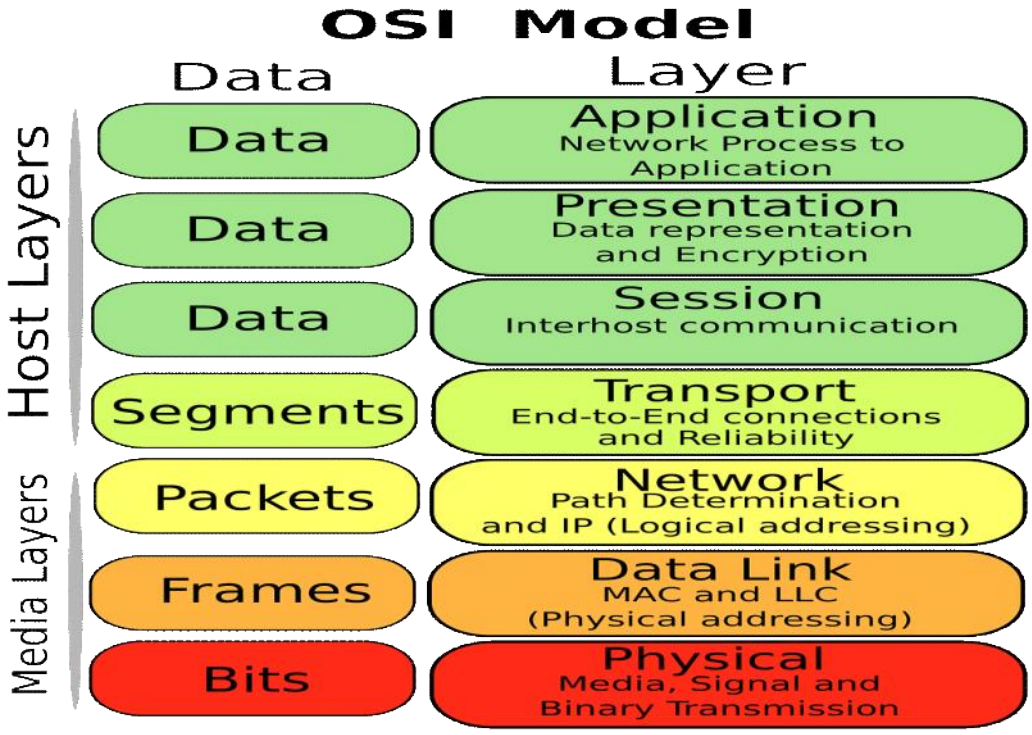
Networking Fundamentals :

Thursday, December 25, 2025 11:59 AM

• OSI Model :

- **The OSI Model is a 7-layer conceptual framework that explains how data moves from one device to another over a network.**
- **It was standardized by the International Organization for Standardization (ISO) to help vendors and engineers design, build, and troubleshoot network systems in a consistent way.**



OSI Layers (Layer 7 → Layer 1) :

◇ Layer 1 – Physical Layer :

- **Transfers raw bits (0s and 1s) over physical media.**

Data Unit:

- **Bits**

Transmission Media

- **Wired:**
  - **Coaxial Cable**
  - **Fiber Optic Cable**
- **Wireless:**
  - **Wi-Fi (Radio Waves)**
  - **Bluetooth**
  - **Infrared**

☑ **Physical layer transmits raw bits over physical media.**

◇ Layer 2 – Data Link Layer:

★ Handles framing, MAC addressing, and error detection.

Data Unit:

★ Frames

Protocols / Standards:

- Ethernet
- ARP
- PPP

Devices:

- Switches
- Network Interface Card (NIC)

☑ Data Link layer provides framing, MAC addressing, and error detection.

◇ Layer 3 – Network Layer :

★ Handles logical addressing and routing between networks.

Data Unit:

★ Packets

Protocols:

- IP (IPv4 / IPv6)
- ICMP
- Routing protocols (OSPF, RIP, BGP)

Devices:

- Routers
- Layer 3 switches

☑ Network layer is responsible for logical addressing and routing.

◇ Layer 4 – Transport Layer

★ Provides end-to-end communication, flow control, segmentation, and error handling.

Data Unit:

★ Segments

Protocols:

- TCP (Reliable, ordered, error-checked)
- UDP (Fast, unreliable)

☑ Transport layer provides reliable or unreliable data delivery using TCP or UDP.

◇ Layer 5 – Session Layer :

★ Controls session establishment, maintenance, synchronization, and termination.

Functions:

- Session setup & teardown
- Checkpointing
- Session recovery

☑ Session layer establishes, manages, and terminates communication sessions.

◇ Layer 6 – Presentation Layer :

★ Handles formatting(Makes data readable and secure for the receiver), encryption/decryption, and compression.

Functions:

- Data translation (ASCII ⇔ Binary)

- **Encryption / Decryption**
- **Compression**

Protocols / Formats:

- **SSL / TLS**
- **JPEG, PNG**
- **MP3, MP4**

☑ **Presentation layer ensures data is properly formatted, encrypted, and compressed.**

◇ Layer 7 – Application Layer :

★ **This is where users and applications interact with the network.**

Protocols:

- **HTTP / HTTPS**
- **FTP**
- **SMTP**
- **DNS**
- **SNMP**

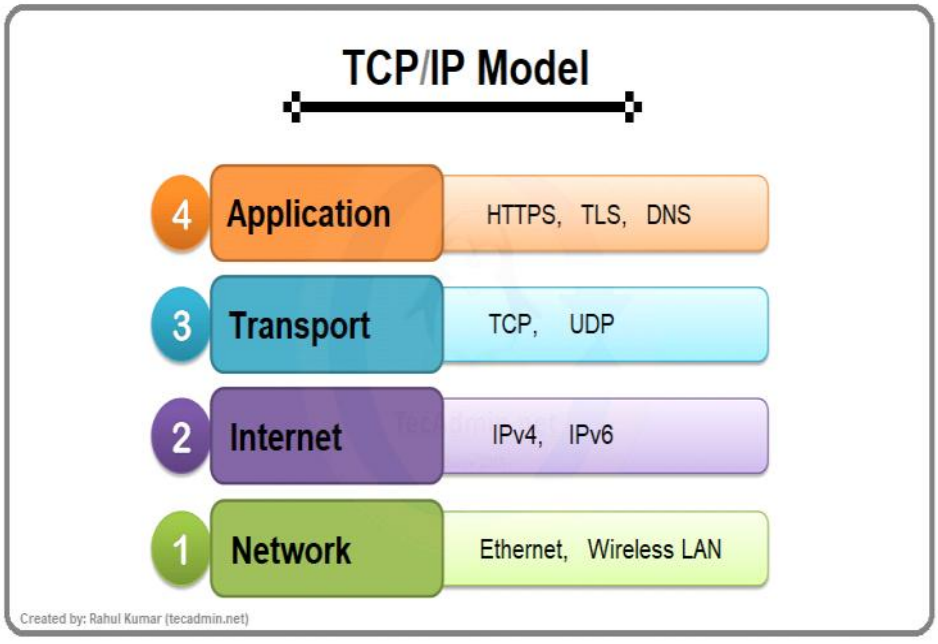
Example:

- **Opening a website in a browser**
- **Sending an email**

☑ **Application layer provides network services directly to end-user applications.**

• TCP/IP :

★ **The TCP/IP (Transmission Control Protocol / Internet Protocol) model is a practical and implementable communication model that defines how data is actually transmitted across networks, and it is the foundation of the Internet.**



★ **TCP/IP HAS 4 LAYERS :**

Application
Transport
Internet

Network Access

◇ 1. Application Layer (TCP/IP) :

- Provides services to user applications like web browsing, email, and file transfer.

Protocols:

- HTTP / HTTPS
- FTP
- SMTP
- POP3 / IMAP
- DNS
- SNMP

☑ Application layer provides network services directly to end-user applications.

◇ 2. Transport Layer (TCP/IP) :

- Provides end-to-end communication, flow control, segmentation, and error handling.

Data Unit:

Segment

Protocols:

- TCP – Reliable, ordered, error-checked
- UDP – Fast, connectionless, no guarantee

☑ Transport layer ensures reliable or fast data delivery using TCP or UDP.

◇ 3. Internet Layer (TCP/IP) :

- Handles logical addressing and routing of packets.

Data Unit:

- Packet

Protocols:

- IP (IPv4 / IPv6)
- ICMP
- IPsec

☑ Internet layer handles logical addressing and routing using IP.

◇ 4. Network Access Layer (TCP/IP) :

- Defines how data is physically transmitted, including framing, MAC addressing, and signals.

Data Units:

- Frame → Bits

Technologies / Media:

- Ethernet
- ARP
- Wi-Fi
- Cables (UTP, Fiber)
- MAC Address

☑ Network Access layer handles physical transmission and MAC addressing.

◇ DNS (Domain Name System):

- DNS is a name resolution system that translates human-readable domain names (like www.google.com) into IP addresses (like 142.250.190.14).
- DNS is the phonebook of the Internet.

◇ Port & Protocol

- Port: 53
- Protocol: UDP (mostly), TCP (large queries)

How DNS Works :

1. User types www.google.com
2. DNS checks cache
3. If not found, queries DNS server
4. DNS returns IP address
5. Browser connects using IP

Name → DNS → IP Address

- DNS Record Types :

Record	Purpose
A	Domain → IPv4
AAAA	Domain → IPv6
CNAME	Alias
MX	Mail server
NS	Name serve

◇ DHCP (Dynamic Host Configuration Protocol):

- DHCP is a network management protocol that automatically assigns IP configuration (IP address, subnet mask, gateway, DNS) to devices.
- DHCP is like a network auto-assigner — it gives devices an IP address automatically.

Port & Protocol :

- Server Port: 67
- Client Port: 68
- Protocol: UDP

How DHCP Works:

1. Discover – Client looks for DHCP server
2. Offer – Server offers IP
3. Request – Client requests IP
4. Acknowledge – Server confirms

What DHCP Assigns:

- IP Address
- Subnet Mask
- Default Gateway
- DNS Server

☑ DHCP automatically assigns IP configuration to network devices.

◇ VPN (Virtual Private Network) :

- A VPN is a secure communication technology that creates an encrypted tunnel over a public

network (like the Internet) to securely transmit data between users and networks.

★ A VPN is a secure tunnel on the Internet that keeps your data private and hidden.

What VPN Does

- Encrypts data
- Hides IP address
- Secures communication over public networks
- Provides remote access

VPN Protocol Examples

- IPsec
- SSL / TLS
- L2TP
- OpenVPN
- WireGuard

☑ A VPN securely encrypts data and provides private communication over public networks.

◇ Firewall :

★ A Firewall is a network security device or software that monitors, filters, and controls incoming and outgoing network traffic based on predefined security rules.

★ firewall can also show configuration details and system logs.

What Firewall Does:

- Allows trusted traffic
- Blocks malicious traffic
- Enforces security policies
- Protects internal network

How Firewall Works :

Traffic → Firewall Rules → Allow / Block

Firewalls are used by both network and security teams:

Network Team :

- Troubleshoots connectivity and network issues
- Checks allowed/blocked traffic
- They Will trouble shoot the issues within the internet.

Security Team :

- Monitors traffic for threats and attacks
- Analyzes logs, alerts, and incidents.
- We can monitor the traffic across the organization network.

★ **TYPES OF FIREWALL:**

a. **Host-Based Firewall**
   - Installed on individual devices (like laptops, servers, desktops).
   - Controls incoming & outgoing packets for that single host.
   - Useful for protecting devices inside the network (where a network firewall can't see everything).
   - Example: Windows Defender Firewall.

b. **Network-Based Firewall**
   - Installed at the network boundary (router, dedicated firewall device).
   - Filters all traffic entering or leaving the entire network.
   - Protects multiple devices inside the internal network.
   - Example: Cisco ASA, Palo Alto Networks firewall.

☑ **Simple Formula:**

- **Host-based firewall** = Guard standing at each house's door.
- **Network-based firewall** = Guard standing at the main gate of the whole colony.

Type	Meaning
Packet Filtering	Checks IP & port
Stateful Firewall	Tracks sessions
Proxy Firewall	Acts as middleman
Next-Gen Firewall (NGFW)	Deep inspection, IDS/IPS

◇ IDS & IPS :

IDS – Intrusion Detection System :

IDS monitors network traffic and detects suspicious or malicious activity, then raises alerts.

Key Points :

• Works like a security camera.
• Only detects and alerts when it sees suspicious or malicious activity and also detects the vernurability and gives an alert.
• It does not block the attack — just reports it.

Example:

- **Multiple wrong login attempts → Alert sent to admin**

☑ Formula:
IDS = Detects & Reports 🚨

IPS – Intrusion Prevention System :

IPS detects malicious activity and takes action to stop it.

Key Points:

• Works like a security guard.
• Not only detects but also takes action (blocks, drops, or stops the malicious traffic).
• Prevents the attack from reaching the system.

Example:

- **Multiple wrong logins → IP blocked automatically**

☑ Formula:
IPS = Detects & Blocks 🛑

Types of Detection in IDS :

1. **Signature-Based Detection :**
   They are already known activities OR known patterns within the internet.
   - **Compares traffic against a internet or database of known attack patterns (signatures).**
   - **Good at detecting known threats, but cannot detect new/unknown attacks.**
   - **Example: Detecting a virus because its signature is already in the database.**
2. **Anomaly-Based Detection :**
   - **They are not known or this patterns are not in the internet.**
   - **Builds a baseline of normal behavior (traffic, user actions, etc.) and raises alerts when something unusual happens.**
   - **Can detect new, unknown, or zero-day attacks.**
   - **But may sometimes raise false alarms (false positives).**

• Ports & Protocols :

◇ What is a Protocol :

- ★ A protocol is a set of rules and standards that defines how data is formatted, transmitted, received, and interpreted between network devices.
- ☑ A protocol defines the rules for communication between network devices.

◇ Internet Protocol (IP) Address :

- ★ An IP address is a unique numerical identifier assigned to a device on a network to enable communication.
- ☑ An IP address uniquely identifies a device on a network.

Types of IP Addresses :

IP addresses are mainly classified into two types:

Private IP Address :

- Used inside a local or private network.
- Not routable on the Internet.
- Not globally unique.
- Can be reused in different private networks.

Private IP Address Ranges :

Class	Range
Class A	10.0.0.0 – 10.255.255.255
Class B	172.16.0.0 – 172.31.255.255
Class C	192.168.0.0 – 192.168.255.255

☑ Classful addressing exists historically, but CIDR is used today.

🔑 Check Your Private IP Address :

- Windows → ipconfig
- Linux / macOS → ifconfig or ip addr

Public IP Address :

- Used on the Internet.
- Globally unique.
- Assigned by an Internet Service Provider (ISP).
- Identifies your network to the outside world.

🔑 Check Your Public IP Address :

- Search in browser: whatismyip.com or what is my IP

Common Internet Protocols :

Protocol	Purpose
HTTP / HTTPS	Web communication
FTP / SFTP	File transfer

SMTP / IMAP / POP3	Email communication
DNS	Converts domain names to IP addresses
TCP / UDP / IP	Core data delivery

◇ What is a Port?

★ A port is a logical numerical identifier used to identify a protocol or service running on a device.

◇ Types of Port Numbers :

Port numbers are managed by Internet Assigned Numbers Authority (IANA).

Port Range	Name	Meaning
0–1023	Well-known ports	Standard services
1024–49151	Registered ports	Applications
49152–65535	Dynamic ports	Temporary use

★ ◇ Most Important Ports & Protocols :

Protocol	Port	Transport	Purpose
HTTP	80	TCP	Web traffic
HTTPS	443	TCP	Secure web
FTP	21	TCP	File transfer
SSH	22	TCP	Secure remote login
Telnet	23	TCP	Remote login (insecure)
★ SMTP	25	TCP	Send email
DNS	53	TCP/UDP	Name resolution
POP3	110	TCP	Receive email
IMAP	143	TCP	Receive email
SNMP	161	UDP	Network monitoring
TFTP	69	UDP	Lightweight file transfer